



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.NOS.6

Represent Rational Numbers and Their Opposites on the Number Line

Lesson 7-2 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period

My Notes

Level 2 Standard



TODAY'S OBJECTIVES

Content Objective: I can place rational numbers, including fractions and decimals, on a number line.

Language Objective: I can explain my reasoning using the words rational number, fraction, decimal, and number line.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rational number

Fraction

Decimal

Number line

Equivalent

Integer

Opposite



Tap any word to see its meaning and a picture.

1

A number you can write as a fraction of two integers, with a bottom number that is not zero, is a .

2

A number that shows part of a whole, like $\frac{3}{5}$, is a .

3

A number that uses a point to show parts of a whole is a

.

4

A line with numbers in order used to place and compare values is a

.

5

Numbers that name the same value, like $\frac{1}{2}$ and 0.5, are

.

6 A whole number or its opposite is an .

7 A number the same distance from 0 on the other side is its
 .

Write About the Math

Put the four readings in order from lowest water to highest. How do the decimals let the keeper say something a whole number could not?

Pon las cuatro lecturas en orden del agua más baja a la más alta. ¿Cómo le permiten los decimales al encargado decir algo que un número entero no podría?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Rational number**
Número racional

☐ **Fraction**
Fracción

☐ **Decimal**
Decimal

☐ **Number line**
Recta numérica

☐ **Equivalent**
Equivalente

Model: $-1 \frac{1}{2}$ falls between ____ and ____ because _____. — Student places $-1 \frac{1}{2}$ between -1 and -2, halfway, reasoning from the value of the fraction part.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The lowest reading was ____ and the highest was ____.** Tuesday's ____ ft means the water sat $1\frac{1}{4}$ feet ____ the mark, which is between -1 and ____ on the number line.
- **My answer is ____ because ____.**
Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Rational number
2. **Notes 2:** Fraction
3. **Notes 3:** Decimal
4. **Notes 4:** Number line
5. **Notes 5:** Equivalent
6. **Notes 6:** Integer
7. **Notes 7:** Opposite
8. **Watch for this mistake:** *A common mistake in Rational Numbers on the Number Line is treating the numerator and denominator of a negative fraction as two separate integers instead of one value between two integers — for example, plotting $-3/4$ by going to -3 and then counting 4 more units to the left, landing (incorrectly) at -7 , instead of recognizing that $-3/4 = -0.75$, which sits between 0 and -1 . Before you submit an answer, find the two integers your value falls between, then split that space into equal parts.*
9. **Try It 1:** A. -1.5 , -0.5 , 0.5 — *The farthest left on the number line comes first: -1.5 , then -0.5 , then 0.5 .*
10. **Try It 2:** C. $-1/4$ — $-3/4 = -0.75$ and $-1/4 = -0.25$. Since -0.75 is farther left (less), $-3/4$ is to the left of $-1/4$.